

Capstone presentation deck

SLIDE 01 / 08 · TITLE

ControlSift

Can a small language model help under-resourced teams tell proof from paperwork?

- MMC · DeepMind AI Research Foundations capstone
- Official SDG: **Goal 10 — Reduced Inequalities**
- Sealed synthetic protocol · not a production auditor

Unequal access to honest assurance

- GRC expertise is unequally distributed
- Policy PDFs get treated as operational proof
- False assurance hits smaller organizations hardest
- Need: open triage research, not autopilot certification

Research aim

- Five-class evidence quality on a sealed synthetic benchmark
- Ladder: majority → TF-IDF → Gemma zero-shot → few-shot → QLoRA
- Primary metric: macro F1 · seed 42 · human judgment remains final
- Hypotheses frozen before LLM claims (protocol-v1-locked)

Method & run setup

- 1,500 compositional packets · family-isolated splits (1000/200/200/100)
- Dataset v1.1.0 after lexical-ceiling harden
- Model: google/gemma-3-1b-it · QLoRA 4-bit NF4 on free-tier GPU
- Malformed generations → UNPARSEABLE (count against score)

Sealed TEST macro F1 (n=200)

- Majority: **0.0667**
- TF-IDF: **0.5327** — still leads (challenge 0.5237)
- Gemma zero-shot: **0.0796** · parse 0.760
- Gemma few-shot: **0.1365** — best Gemma rung · parse 0.985
- Gemma QLoRA: **0.0827** · parse **0.435** — did not beat few-shot

Responsible innovation

- Synthetic data only · Data Card · Risk Register · Intended Use
- No production-auditor claims · no fabricated metrics
- Failure Lab documents hard cases (SUFFICIENT / PARTIAL / IRRELEVANT)
- Preserve negative findings: QLoRA parse fragility is a result

Impact (SDG 10)

- Open methods reduce dependence on expensive GRC consulting
- Free-tier reproduction keeps access wide
- Triage aid for human reviewers — not automated certification
- Honest reporting models how under-resourced teams can still publish rigor

Status & ask

- Research metrics, PDFs, badge, and GitHub Pages are **done**
- Remaining: record ≤5-minute video · upload via GCLP form
- Artifacts: research site · report · slides · Assurance · curriculum map
- Thank you — questions welcome

